

SIDERIUS

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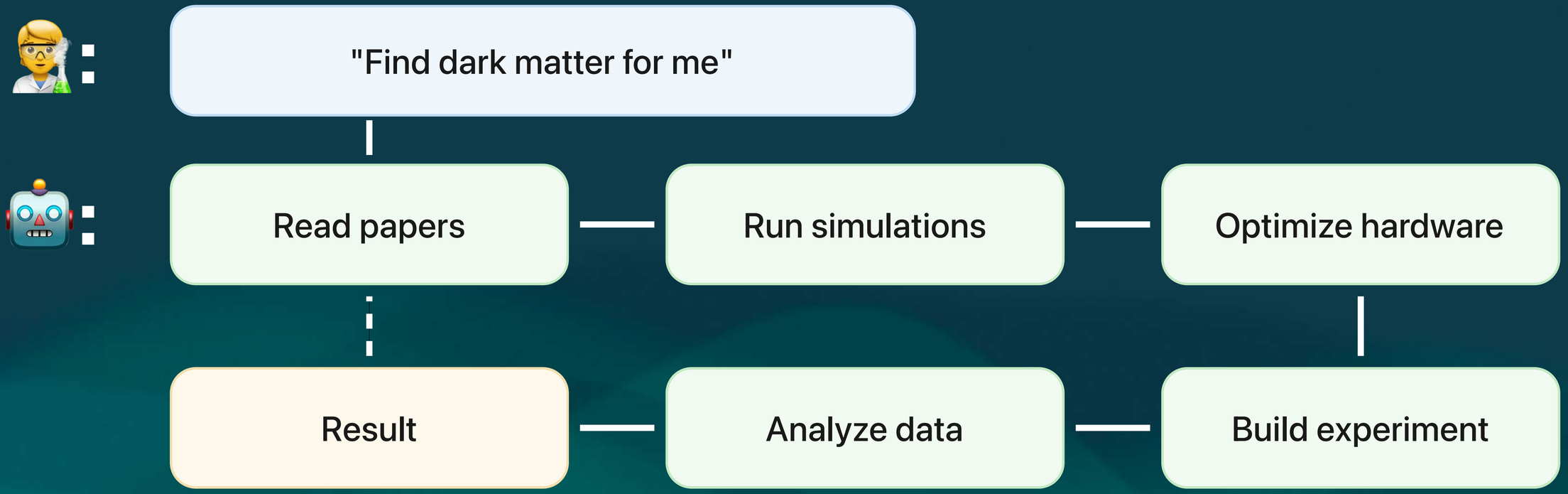
Resource-Aware Multi-Agent System for Automated Machine Learning in Science

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20260902@Fast Machine Learning for Science Conference 2026

Introduction: Background of Science Agents

The Ultimate Goal, a Big Dream



Good. Only ping me when you really find it

A Few
~~Moments~~ ~~Years~~ Decades
Later…

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The Ultimate Goal, a Big Dream

🤖: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

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Two Key Words in this Dream:

Introduction: Background of Science Agents

The Ultimate Goal, a Big Dream

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Two Key Words in this Dream:

End-to-end: Agents handles every stage, automates the loop

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🤖: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Two Key Words in this Dream:

End-to-end: Agents handles every stage, automates the loop

Long-horizon: Agents keep working for days, months, years

Introduction: Background of Science Agents

Yes, but…

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Case 1

🤖: Look at this awesome Axion that I found!
According to literature,
It is one of the dark matter candidates



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Problem with this Dream:

Drifting/Gaming: When the goal is not clear (and it is always hard to have a 100% guarantee!), the agent optimizes endlessly, but chasing a phantom

Introduction: Background of Science Agents

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Case 2

🤖: Now the main problem is solved!
One last minor thing that I need you to do:
pay the 1 trillion bill for token usage, GPU computation, and instrumentation.

Introduction: Background of Science Agents

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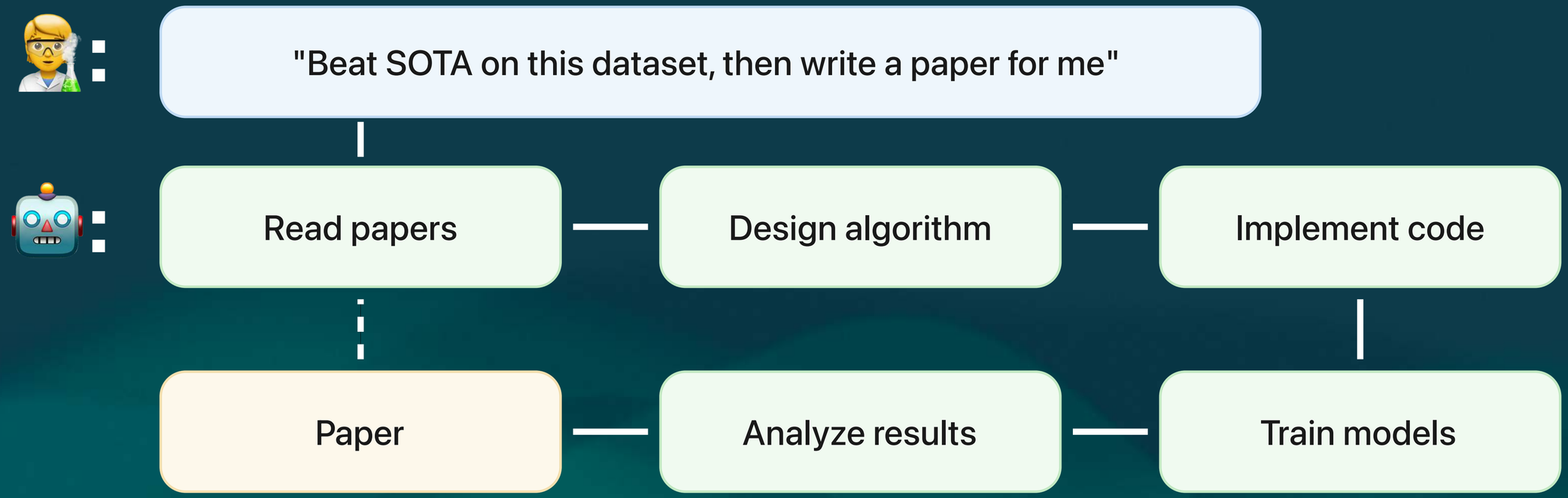
🤖: Hey, come to have a look! I got a 5 sigma discovery with Experiment 289421, Dark Matter Candidate 320382

Problem with this Dream:

Wasting: Coming up with a huge amount of hypotheses is easy for LLM, verifying them is not. **Science is not simply brute-force repeating.**

Zoom-in: Automatic ML in Science

An Easier Task (Not Finding Dark Matter)



This is already happening.

The easiest case:
send “/goal Beat SOTA on this dataset, then write a paper for me” to codex

Zoom-in: Automatic ML in Science

An Easier Task (Not Finding Dark Matter)

Drifting/Gaming: Overfit, Mode Collapse, etc

Zoom-in: Automatic ML in Science

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Human-in-the-loop: adjust the road map; set guardrails

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Wasting: GPU is expensive, data storage is expensive too

Zoom-in: Automatic ML in Science

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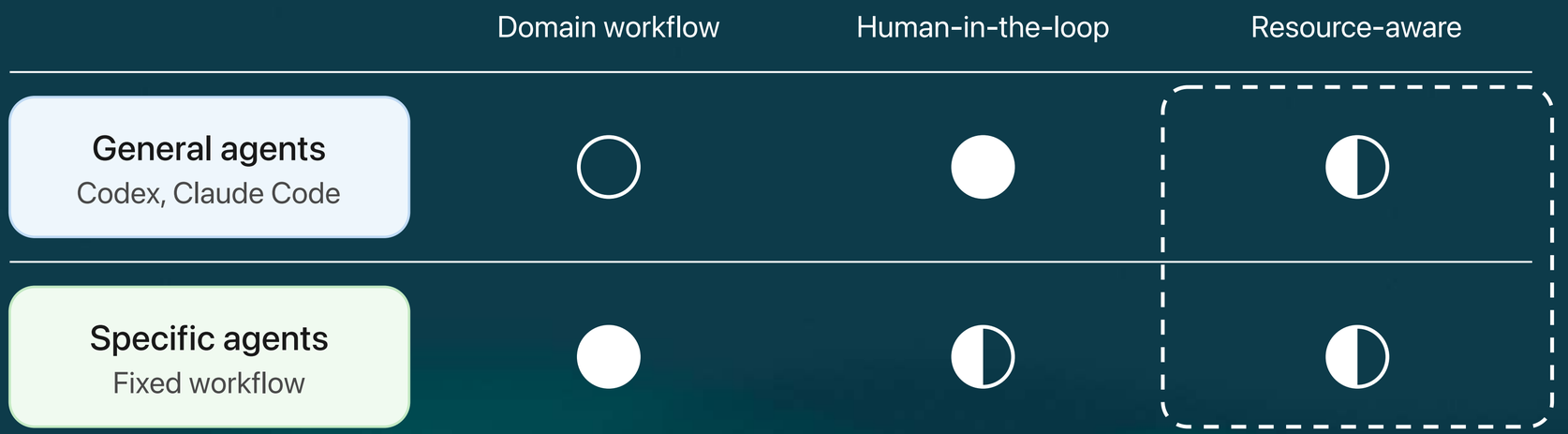
Wasting: GPU is expensive, data storage is expensive too

Resource-awareness:

- Predict before execution
- Plan with the budget
- Reflect on the cost

Zoom-in: Automatic ML in Science

The current status, and gaps



Can we build something in between, and better?

The motivation of SIDERIUS

- **Human-in-the-loop**

  Trade-off between flexibility and restriction

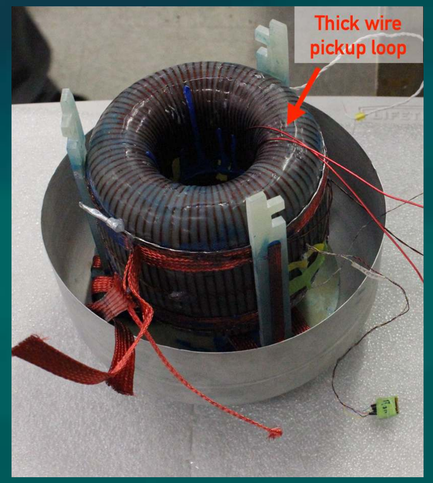
- **Resource-awareness**

  LLM does a very bad guessing, unless we teach it explicitly

The Background of SIDERIUS

TIDMAD: A Sandbox for LLM Agent to Search for Dark Matter





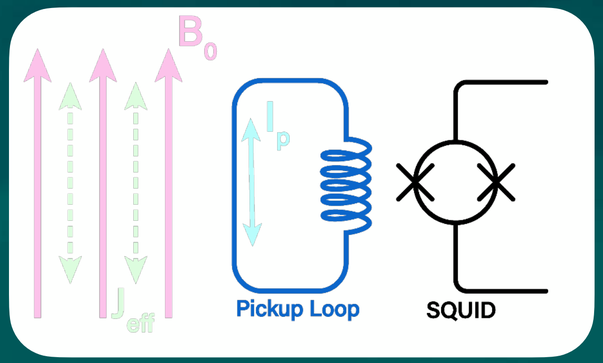
Axion-Modified Maxwell's Equation

at $L \ll \lambda_a$ **:**

$$\nabla \times \mathbf{B} \approx \mathbf{J}_{eff}$$

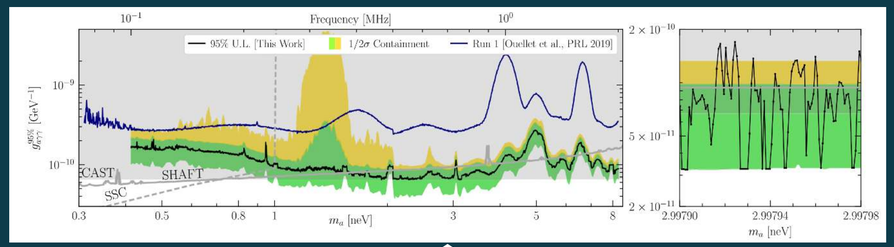
$$J_{eff} = g_{a\gamma\gamma}\sqrt{2\rho_{DM}}cos(m_a t)B$$

Magnetic Flux created by the effective current



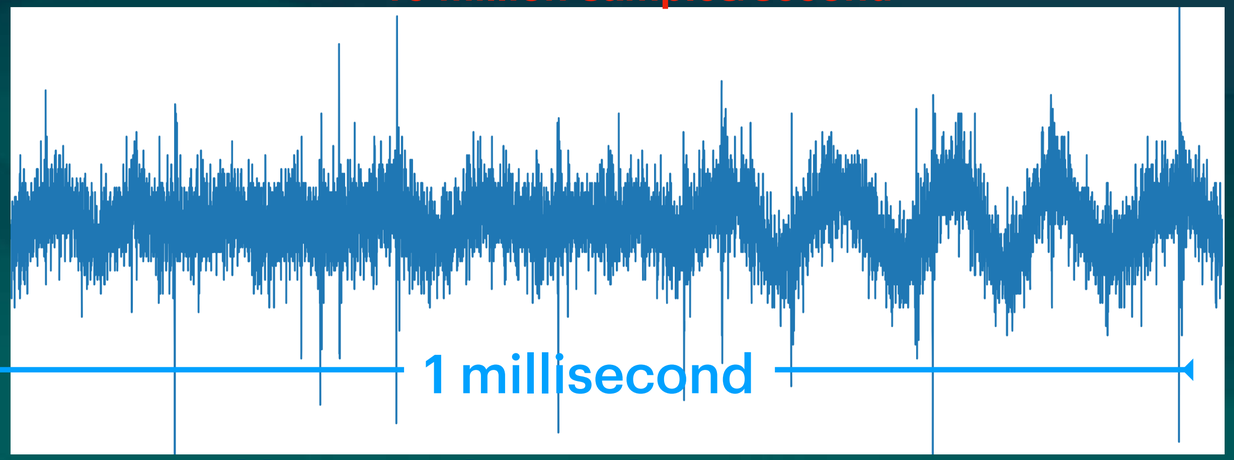
Frequency Spectrum

Broadband search for axion DM



Ultra-long Time Series

10 million samples/second

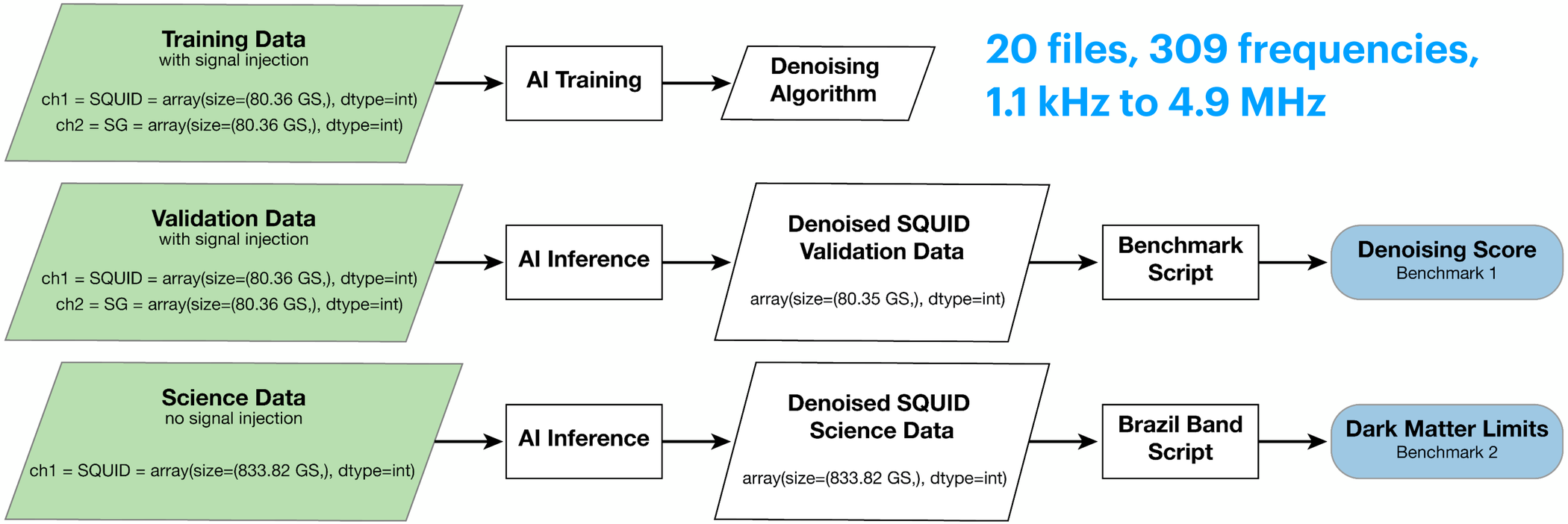


Experimental Apparatus Constructed by Winslow Lab at MIT

The Background of SIDERIUS

TIDMAD: A Sandbox for LLM Agent to Search for Dark Matter

Labeled Pair: [Signal only] — [Signal + noise]

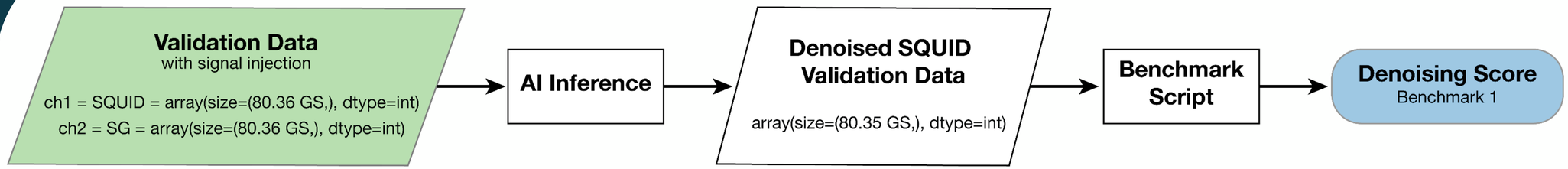


20 files, 309 frequencies,
1.1 kHz to 4.9 MHz

Search for dark matter in AI-denoised data

The Background of SIDERIUS

The Metric to Optimize in TIDMAD



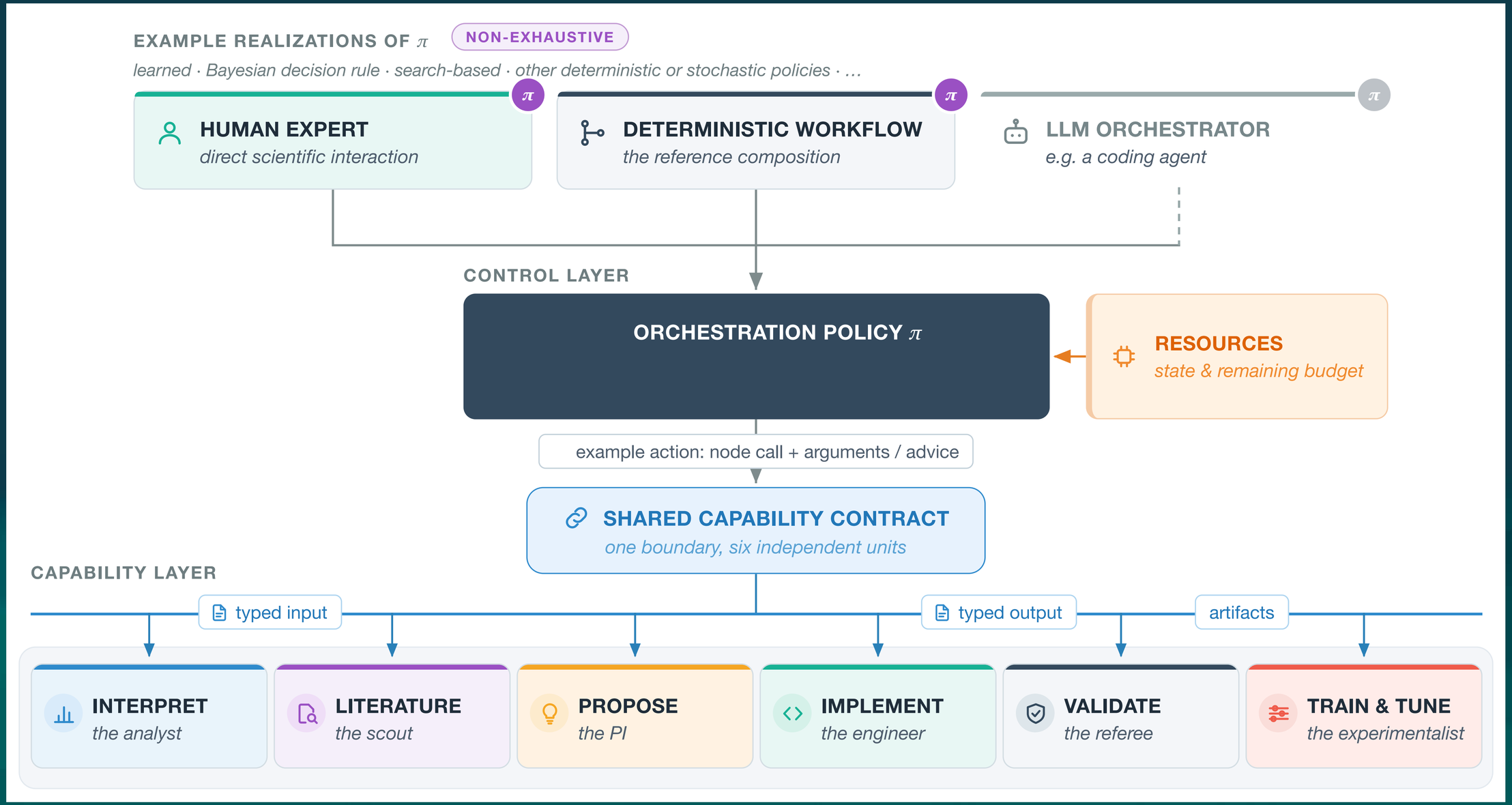
Motivation: Axion signal is a monochromatic wave

Denoising Score: Signal-to-noise Ratio in the frequency spectrum

$$\Lambda = \left(\frac{1}{n} \sum_{i=0}^{n} (SNR_{\mathrm{SQUID}})_i \times (SNR'_{\mathrm{Injected}})_i \right)$$

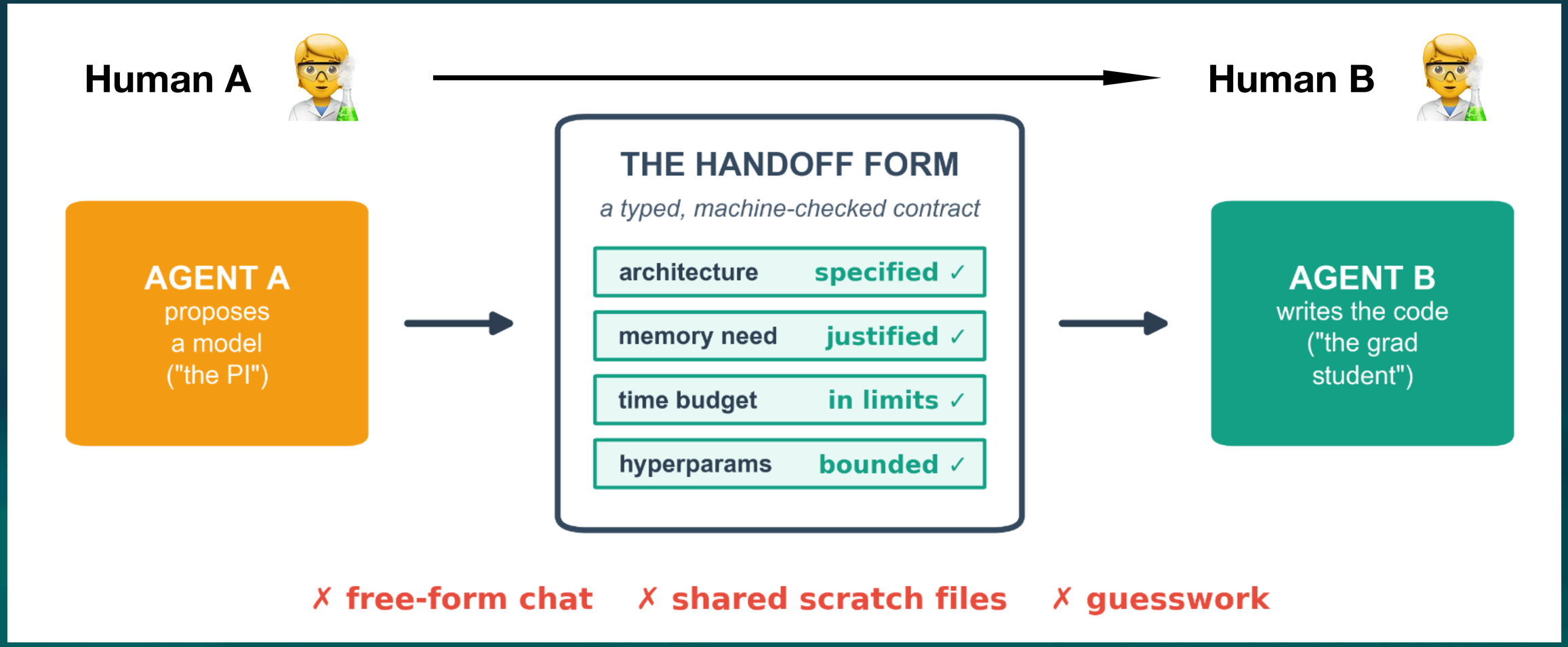
$$\text{Denoising Score} = log_{5.27}\Lambda$$

SIDERIUS: The Full Structure



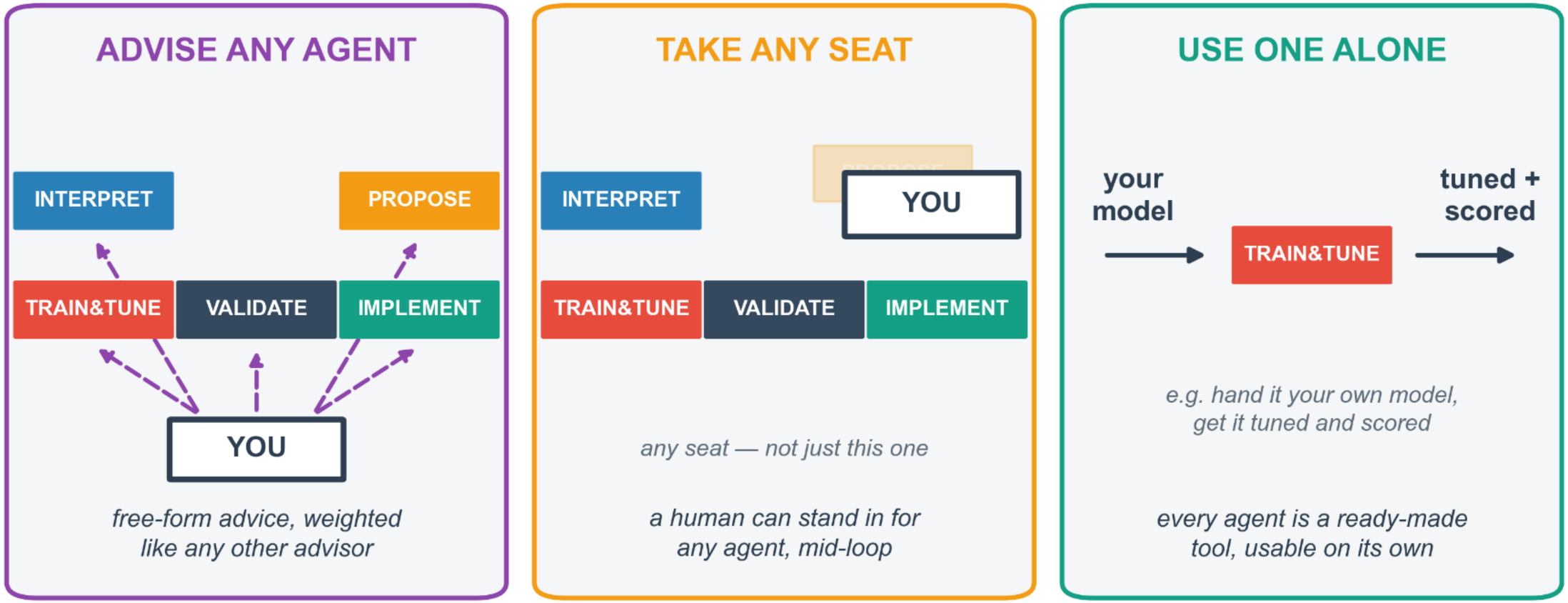
SIDERIUS: The Design Principle

Contract-based, inspired by functional programming



SIDERIUS: Human-in-the-loop

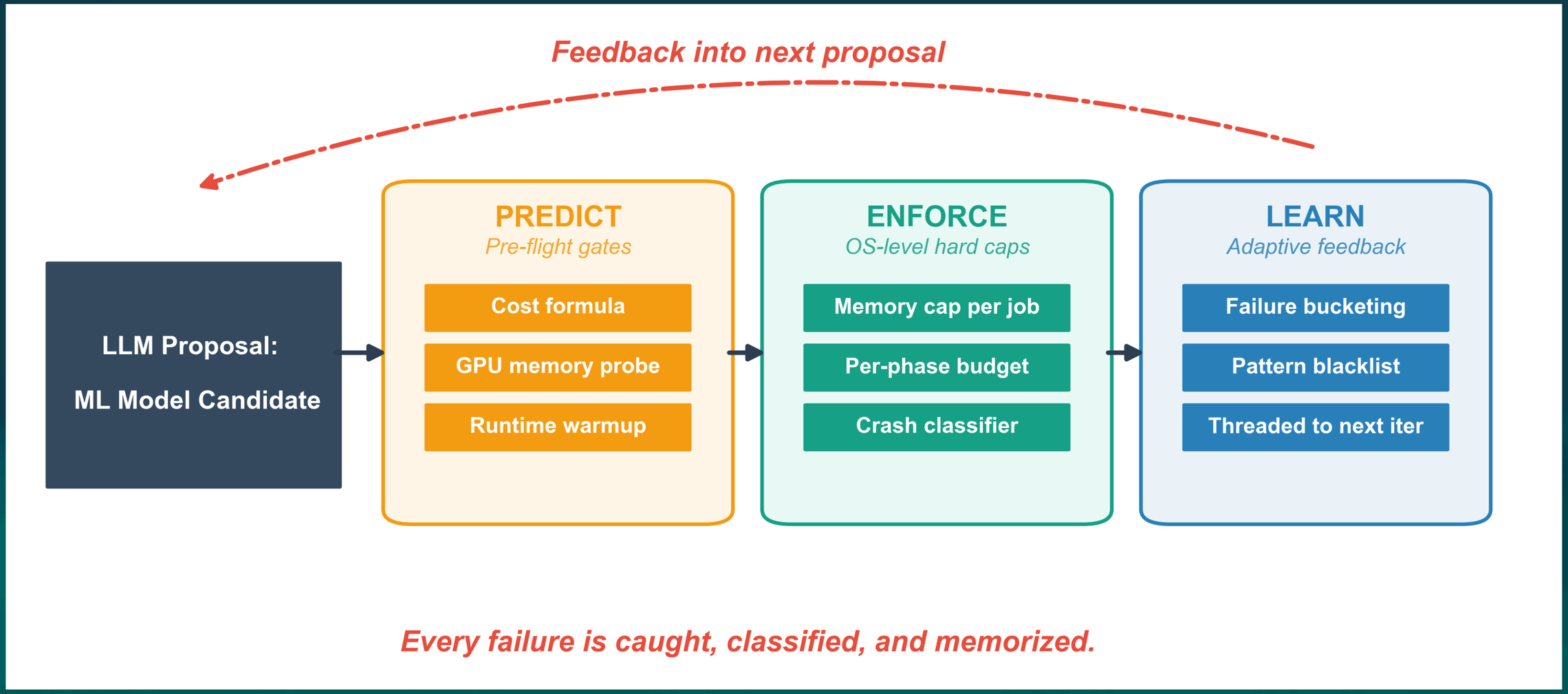
Agents are tools, use them in anyway you like



One typed contract for every seat — humans and agents are interchangeable.

SIDERIUS: Resource-awareness

Multi-layer of Resource-awareness



Every failure is caught, classified, and memorized.

What We Found: Evaluation Matters

Real-world science is much beyond one single metric

Denoising Score: Signal-to-noise Ratio in the frequency spectrum

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🤖: This is easy! Let me eliminate signal and noise altogether, so that they are comparable!

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A read hacking mode we found for TIDMAD

- 99.99% of the outputs are 0
- A few non-zero points contribute to the signal frequency

What We Found: Evaluation Matters

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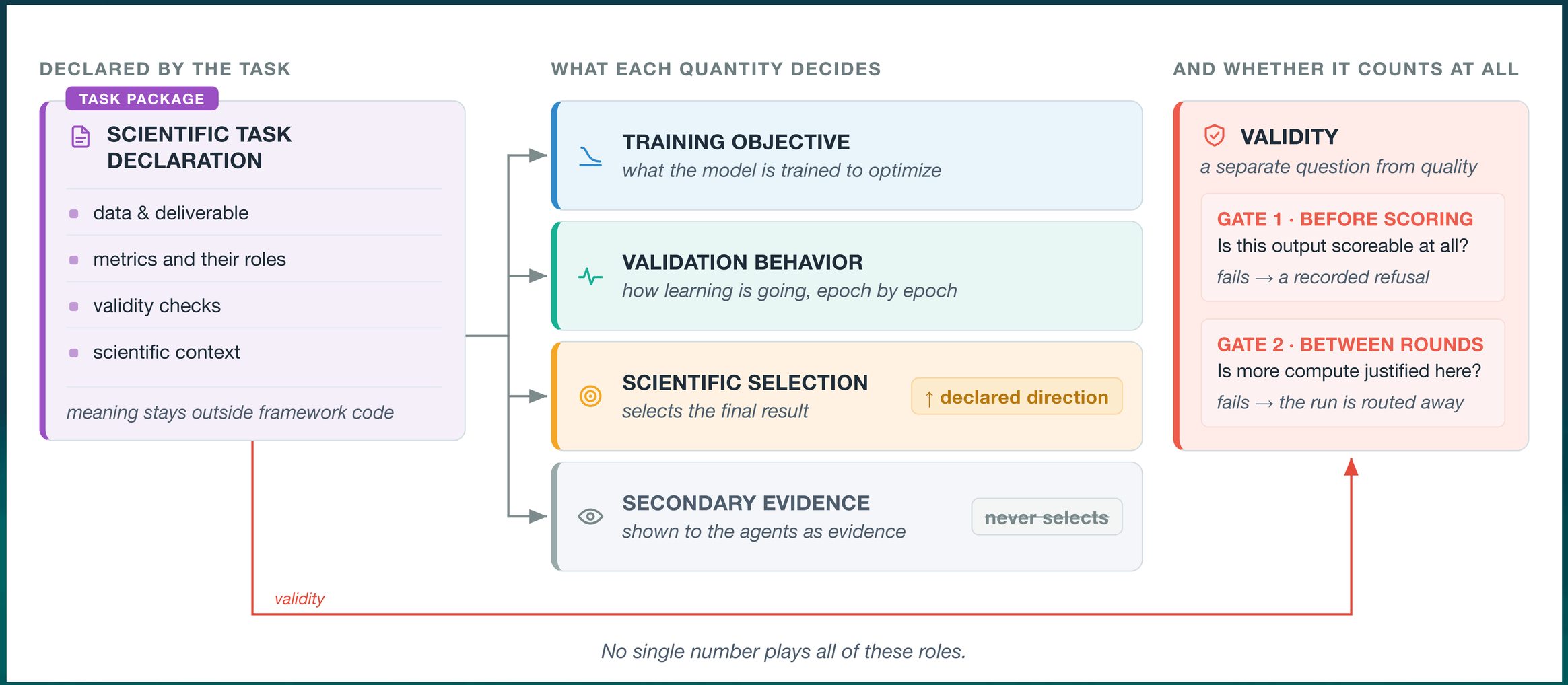
A read hacking mode we found for TIDMAD

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A clear mode collapse here, more stealthy hackings anywhere

Evaluation Matters

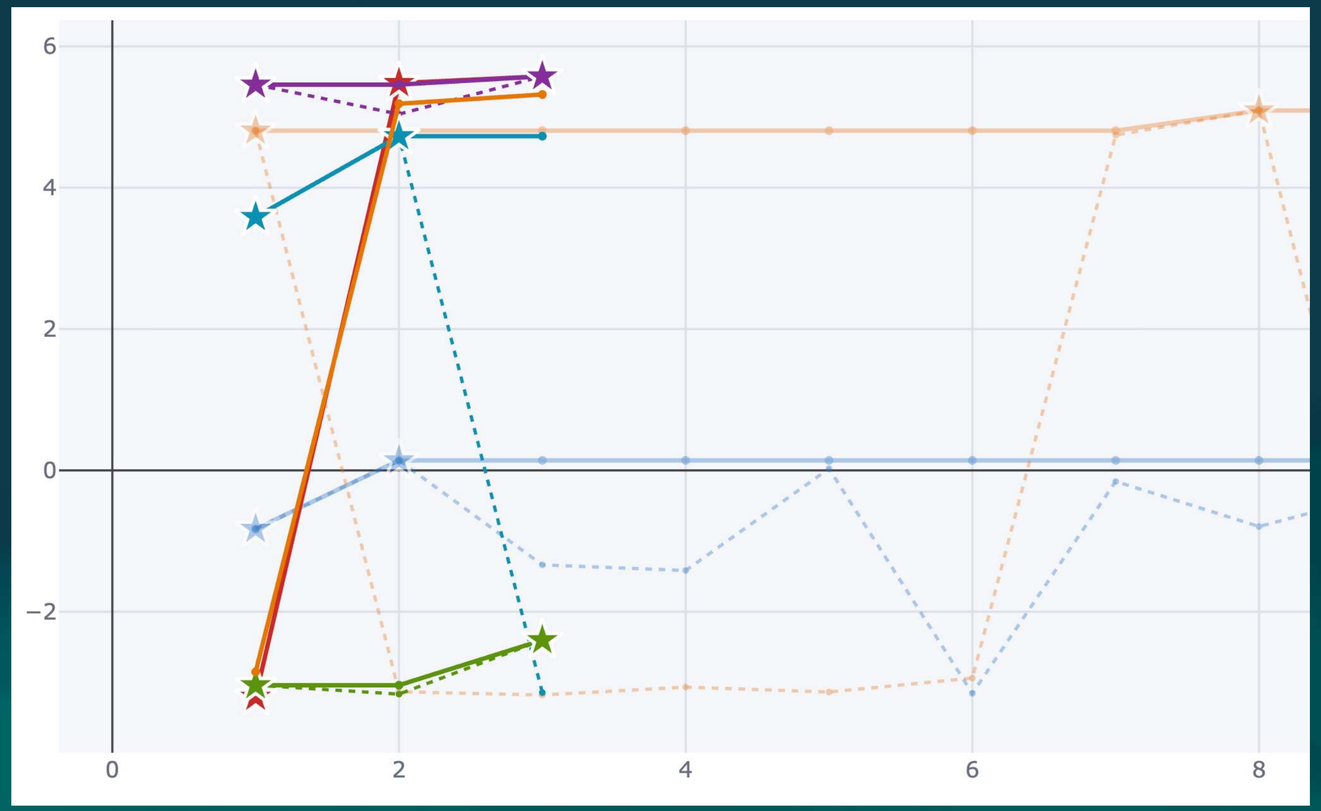
A science task come as a full-suite declaration, not a single goal



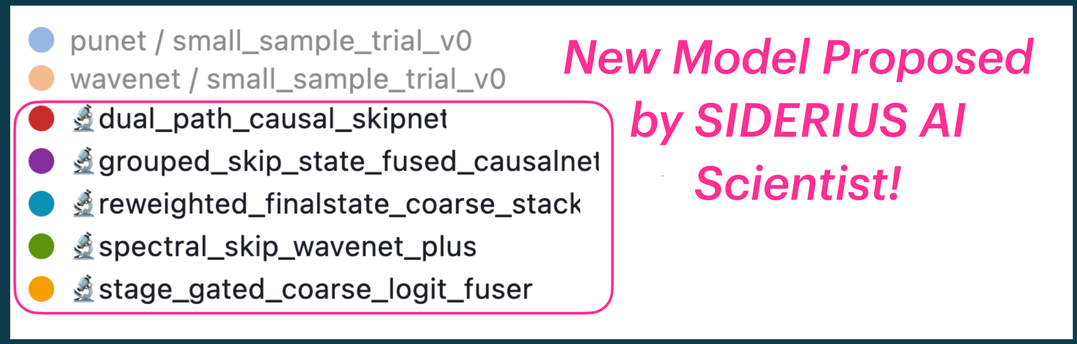
No single number plays all of these roles.

SIDERIUS: Current Progress

Case study: Denoising the Time Series in TIDMAD



Hyper Parameter Tuning Loop



New Model Proposed by SIDERIUS AI Scientist!

Working example*:

- **Efficient new-ML model proposing, training and tuning**
- **Beat previous paper baselines within a few explorations**

*on a single TIDMAD file out of 20

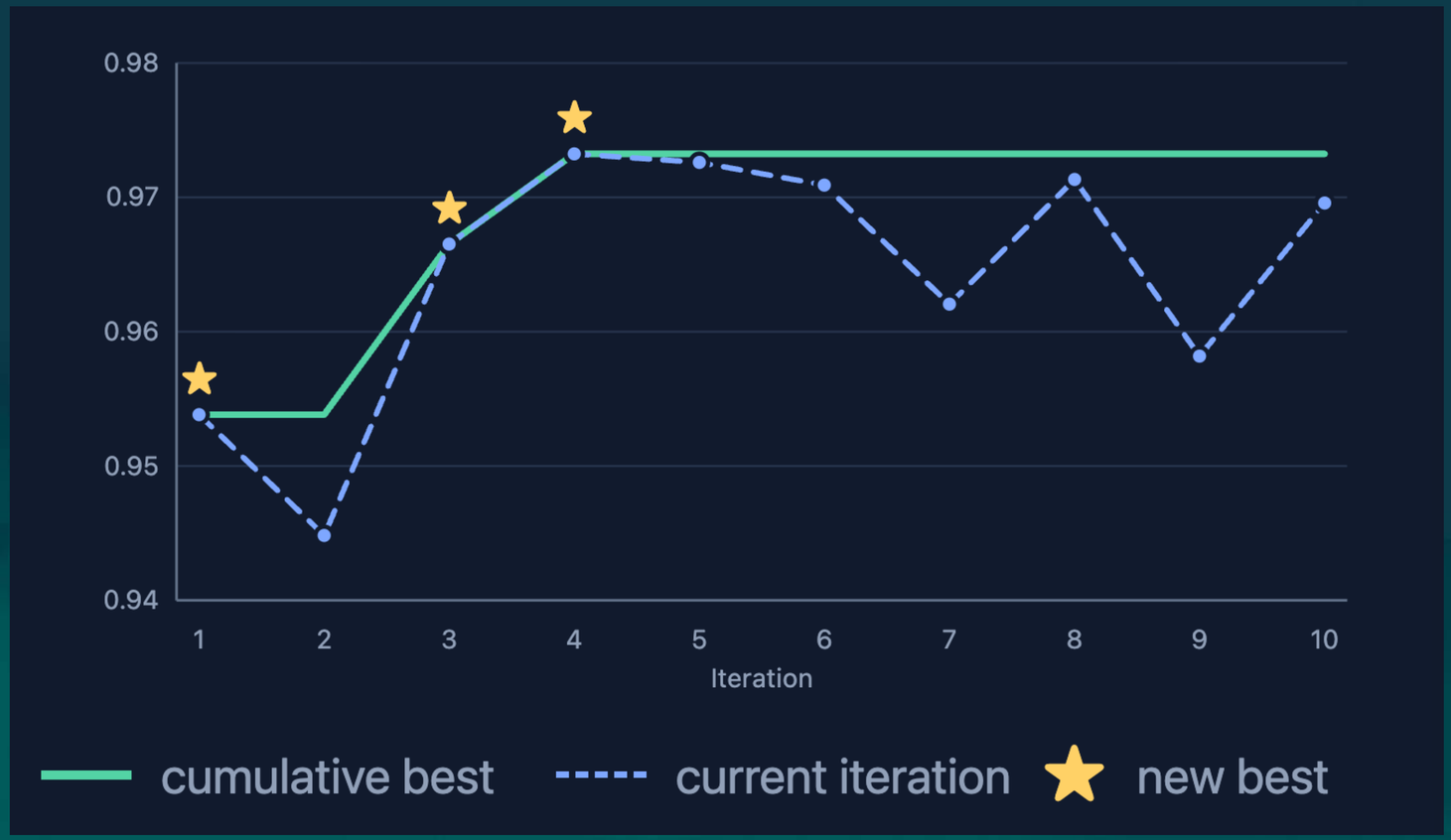
SIDERIUS: Current Progress

Case study: Signal/Background Classification in MAJORANA

Signal: Single Site
Background: Multiple Site

Optimizing for:
classification AUC

- 1 x 5090
- One command
- within 3 hours



SIDERIUS: Current Progress

Case study: Signal/Background Classification in SuperNEMO

Signal:

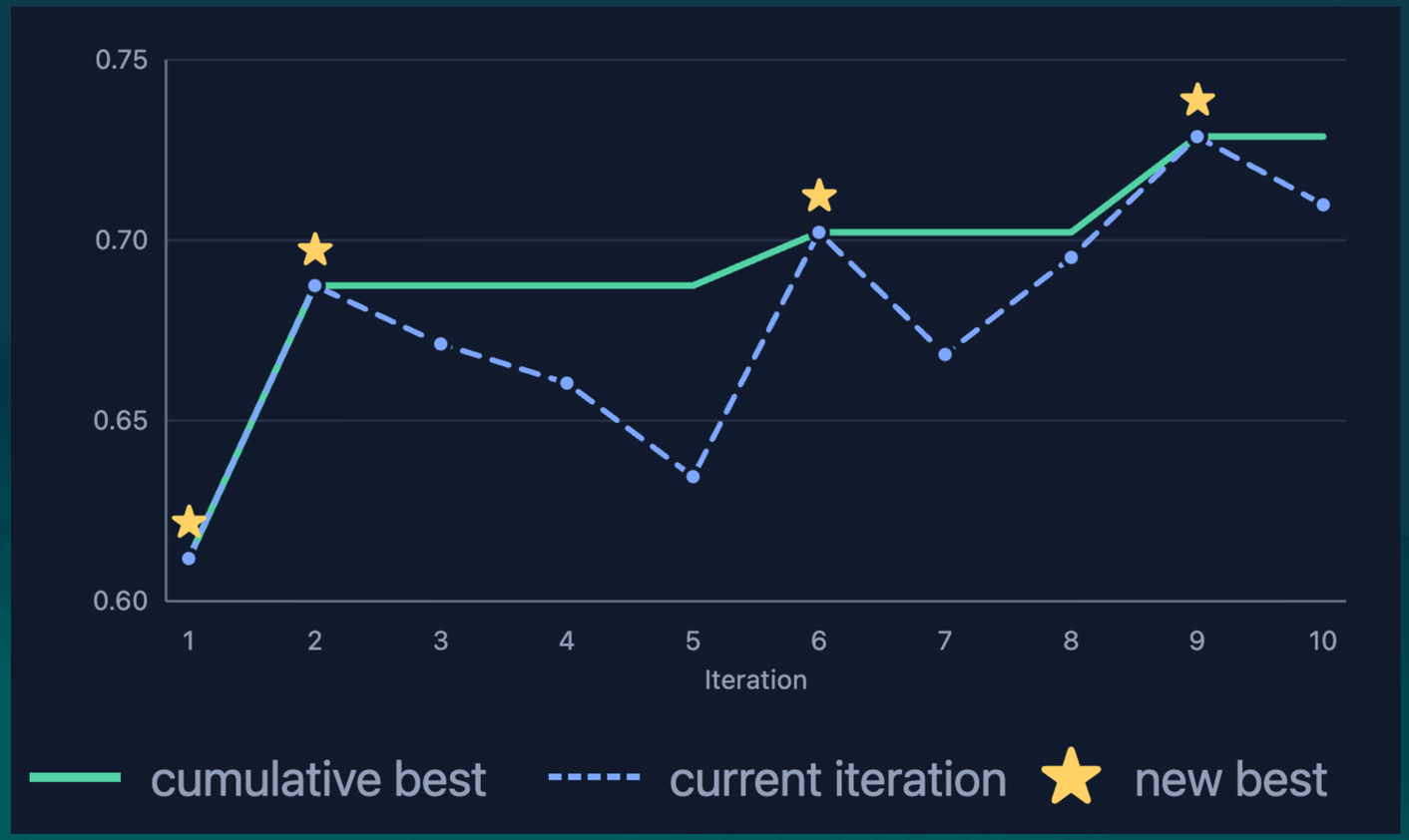
$0\nu\beta\beta$

Background:

$2\nu\beta\beta$, $\beta + [\gamma \rightarrow Compton]$, **etc**

Optimizing for:
classification AUC

- 1 x 5090
- One command
- within 3 hours



SIDERIUS: Prospects

Ongoing Genesis Mission Program Tasks

	AI TASK	DATA	MAIN CHALLENGE
TIDMAD search for axion dark matter	Denoise the series to recover the signal	SQUID magnetometer 0.8 TB per 24 hours	Noise is complex and varies in time
CUPID TES search for 0vbb decay	Denoise the pulses to sharpen the timing	Heat pulses in a cryogenic crystal	Pile-up background fakes the signal
QEC improve quantum computing	Denoise transient signals in real time	Error-check readouts from the qubits	Must finish within about 1 millisecond

SIDERIUS: What to use for/How to use

Call for internal users and collaborators

- Available data format:
  - Time Series/Image/Video/Point Cloud/etc
- Task type:
  - Classification/Regression/etc

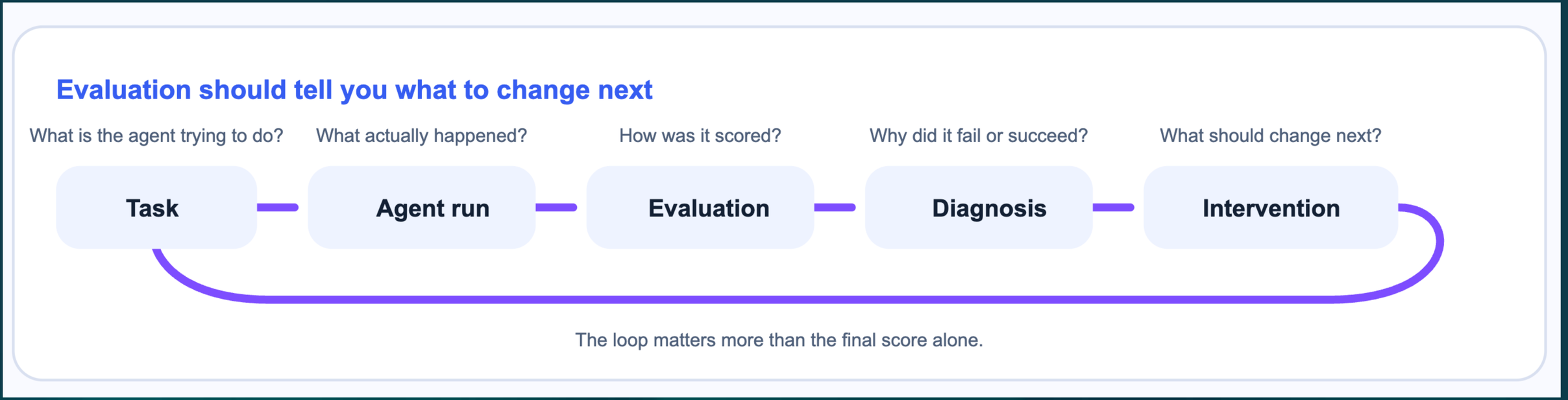
All verified by concrete examples ✅

Minimal Example: Hey Claude/Codex, beat SOTA on this dataset, then write a paper for me. + Use the SIDERIUS infra to do that!

Now - Oct: Open for internal usage. Contact us!
Nov: release SIDERIUS as an open-source framework

Echo: Evaluation Matters

Not only for the task, but also for the agent system itself



Drifting/Gaming: Evaluate the failure patterns, not just the success

Wasting: Evaluate the efficiency, not just the outcome

A Live knowledgebase of LLM agent evaluation: https://yuema137.github.io/scieval/

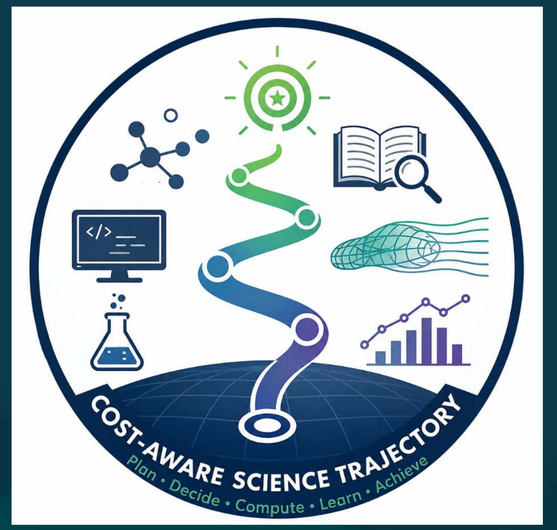
Echo: Evaluation Matters

Join the Effort! All Scientists are Welcome

SciTra

Evaluate Agent **Efficiency** in Cost-sensitive Science/Engineer Task

Build a community for Scientists-AI Co-evolution

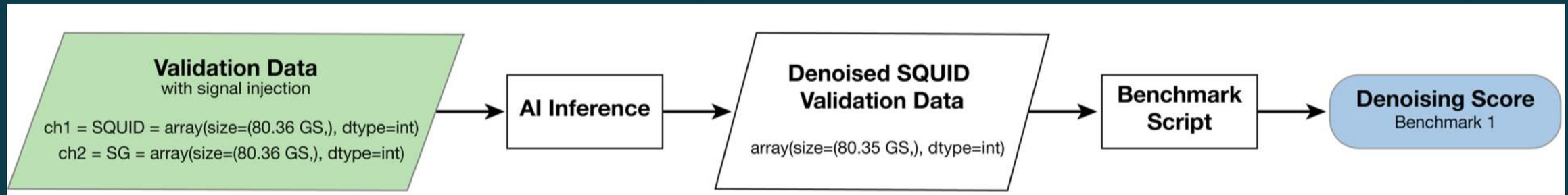


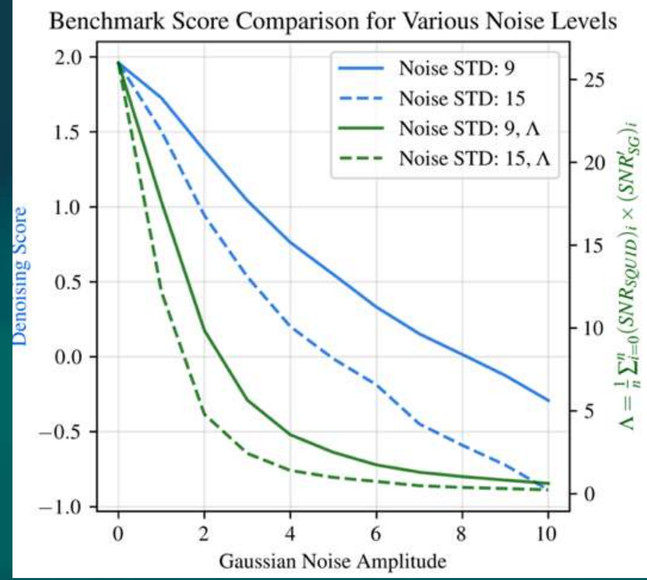
scitra.org

Backups

The Background of SIDERIUS

The Metric to Optimize in TIDMAD

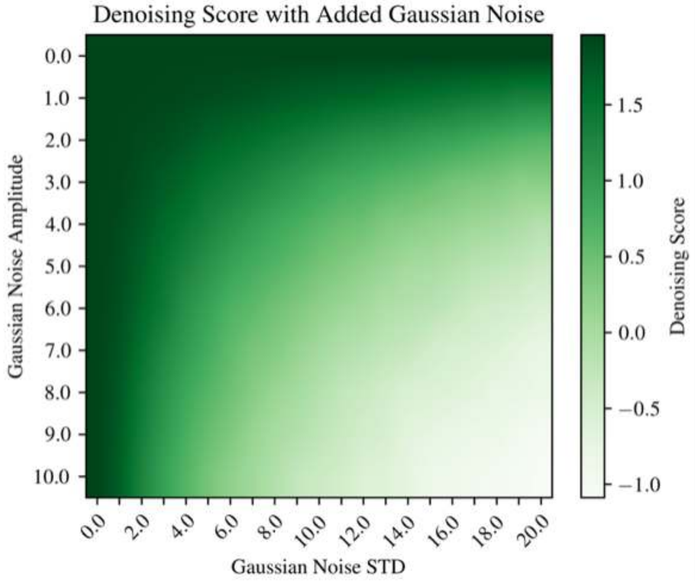




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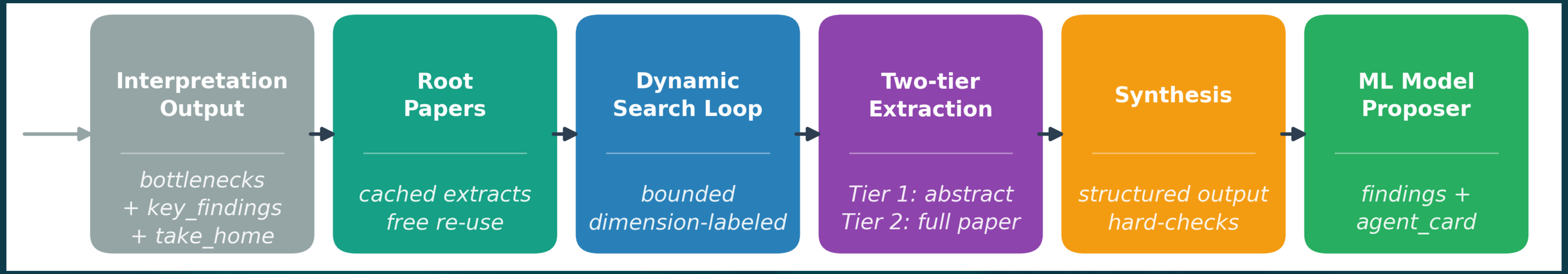
$$\text{Denoising Score} = log_{5.27}\Lambda$$

Test the denoising score by doping gaussian noise into Time Series



SIDERIUS: Literature Review

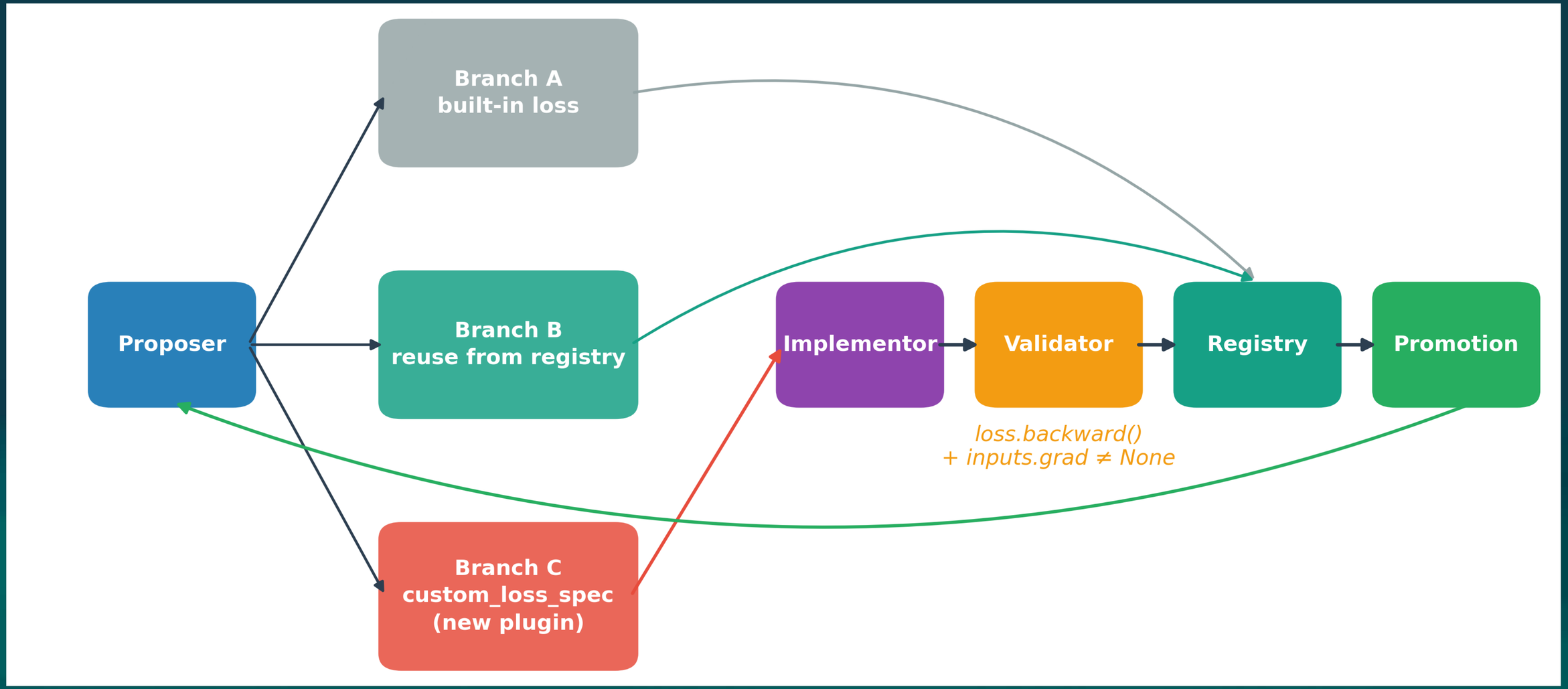
Searching is non-trivial



- **2-Layer Source**: Root paper + Dynamic search
- Search: max round + detention labeled (different aspects)
- **2-Tier Extraction**: Dynamic escalation based on correlation to task
- Protocol to proposer: Synthesis -> finding with confidence + agent_card

SIDERIUS: Toolkit Accumulation

Loss a an example



SciTra: Cost-Aware Science Trajectory

The Bottleneck of Scaling-up: Resource

- **End-to-end research workflow is easy to build**
- **But scientific discovery is fundamentally resource-constrained**

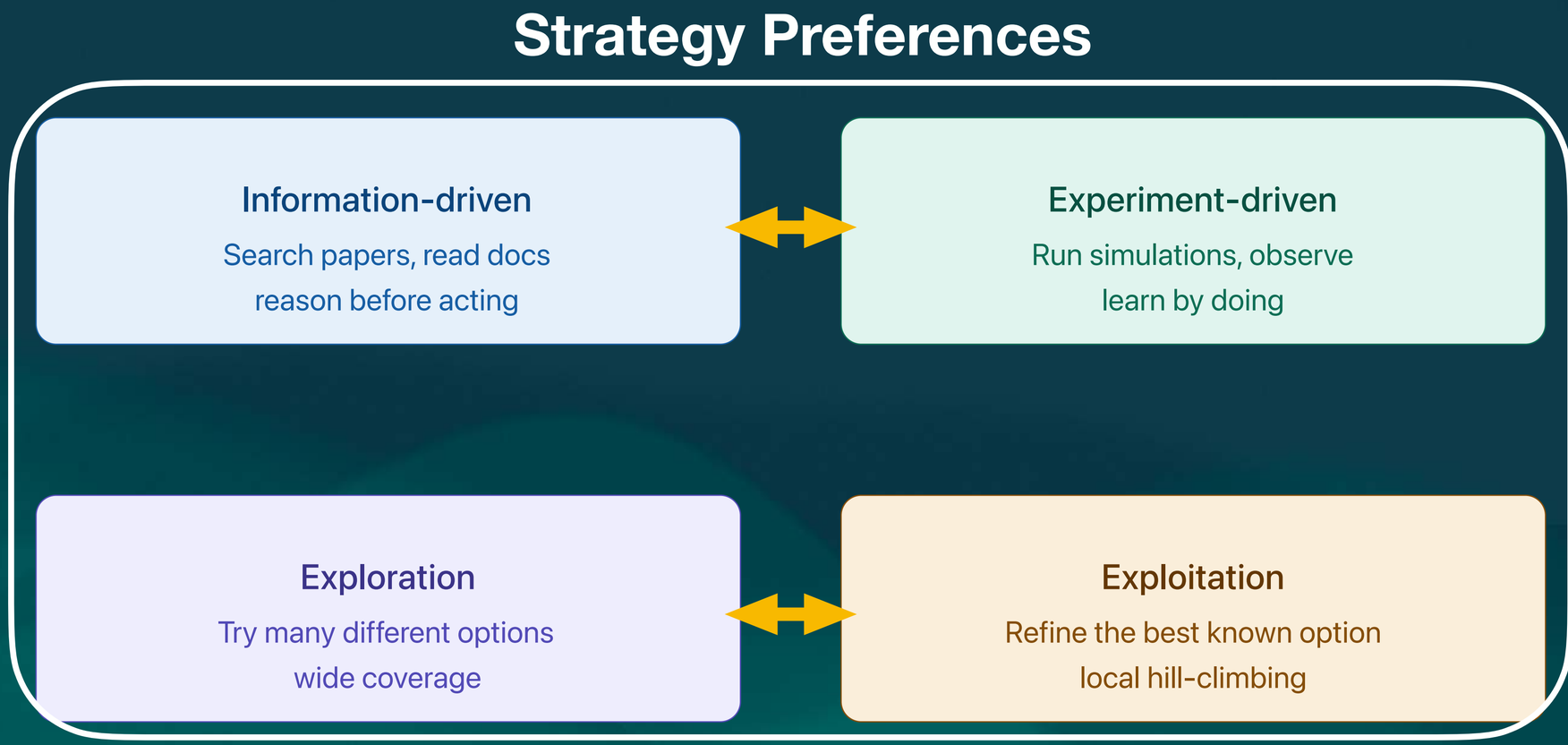


SciTra: Cost-Aware Science Trajectory

From pass@k, to the efficiency evaluation

- **A minimal example:**

👨: Design a car for me, make sure the air resistance is below X at speed of Y



Which is better in terms of efficiency?

Honest answer: agent doesn’t know

SciTra: Cost-Aware Science Trajectory

Whole lifecycle of task collection and evaluation



Trajectory and strategy representation: granularity matters

Coarse:



expand

Fine:

